

12 (**long**) Figure 9 shows a schematic diagram of a vibration transducer consisting of a body of mass m connected to an outer casing through a spring-dashpot arrangement. The absolute displacements of the body and the outer casing are x and y respectively and x is measured with respect to the equilibrium position of the body. The relative displacement of the body with respect to the casing is given by $z = x - y$. An initial choice of device parameters is specified as $m = 1 \text{ kg}$, $k = 100 \text{ N m}^{-1}$, $\lambda = 5 \text{ N s m}^{-1}$.

(a) Show that the equation of motion for this system can be written in the form

$$m \frac{d^2 z}{dt^2} + \alpha \lambda \frac{dz}{dt} + \beta k z = -m \frac{d^2 y}{dt^2}$$

and obtain values for α and β . [6]

(b) Steady sinusoidal motion of the base $y = Y \cos \omega t$ gives rise to a sinusoidal response of the body given by $z = Z \cos(\omega t - \phi)$. Starting from the equation of motion, derive a relationship between Z and Y as a function of ω . Calculate Z and ϕ when $Y = 10 \text{ mm}$ and $\omega = 5 \text{ rad s}^{-1}$. [8]

(c) The device is now configured as a seismic vibration transducer so that the displacement of the body tracks that of the casing. Specify a new value for λ so as to limit the response at resonance to within 40% of the nominal value while maximising the working range of frequencies for device operation. Estimate the damped natural frequency for this value of λ . [6]

(d) The device is dropped on the floor. The resulting acceleration can be modelled as an impulse imparted to the outer casing of unit magnitude $\delta(t)$. Calculate the maximum resulting displacement of the body for the choice of parameters in (c) above. How long does it take for the displacement of the body to settle to within 0.1% of its maximum value? [10]

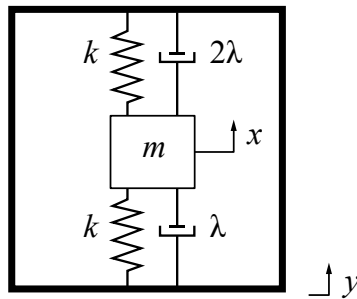


Fig. 9

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Answers

- 1 (a) $\frac{m_A T_A + m_B T_B}{m_A + m_B}$ (b) $T_F = (T_A)^r (T_B)^{1-r}$
- 2 (b)
 - (i) Constant h is also constant T
 - (ii) $(\partial T / \partial s)_p = T / c_p$
 - (iii) $(\partial T / \partial s)_v = T / c_v$
 - (iv) Reversible and adiabatic is also constant s
- 3 (a) 0.00624 m^3 (b) 552.7 K , 417.5 J , 1200.8 J
- 4 (a) 3 (b) $\frac{w}{U^2} = f\left(\frac{V_1}{U}, \frac{U}{\sqrt{c_p T_1}}, \frac{c_p}{R}\right)$ (These are the conventional groups)
- 5 (a) 467 kg (b) Bottom of the gate. $1.6 \times 10^{-4} \text{ m}^2$
- 6 (a) $V_1 = V_1$ (Bernoulli) (b) $A_2 = A_1$ (mass continuity) (d) $F = 2\rho A_1 (V_1 - U)^2$
- 7 (a) MR^2 (b) $\approx 2 \text{ W}$
- 8 (a) $u(\theta) = \sqrt{v^2 - 2Rg(1 + \cos \theta)}$ (b) $\sqrt{5Rg}$
- 9 (a) $\frac{4}{3}\rho\pi Gmr$ (b) $\frac{2}{3}\rho\pi Gm(R^2 - r^2)$ (c) $\frac{2}{9}(\rho\pi a)^2 GR^3$ (d) $\sqrt{4\rho\pi G/3}$
- 10 (a) $\vec{v} = -200\hat{e}_\theta, \vec{a} = \frac{10}{\sqrt{2}}\hat{e}_r - \frac{10}{\sqrt{2}}\hat{e}_\theta, \ddot{r} = 47.07 \text{ ms}^{-1}$ (b) 5.66 km
- 11 (a) Mode shapes: $[0, 1], [0, 1]$, natural frequencies $\omega_x = \sqrt{2k/m}, \omega_y = 4\sqrt{k/m}$
 (b) $x = 0, y = \frac{1}{4}\sqrt{m/k} \sin \omega_y t$
- 12 (a) $\alpha = 3, \beta = 2$ (b) $1.31 \text{ mm}, 0.4 \text{ rad}$ (c) $3.67 \text{ N s m}^{-1}, 16.9 \text{ rad s}^{-1}$
 (d) $43 \text{ mm}, 1.25 \text{ s}$